

File Handling & Pointers in C

Introduction

- ↳ In C, variables store data only while the program is running. Once the program ends, the data is lost. To store data permanently, C provides file handling. Another important concept is Pointers, which store the memory address of variables instead of their values. Both concepts are widely used in real-world applications and form the foundation of advanced programming.

File Handling in C

- ↳ A file is a collection of data stored permanently on a storage device. It allows data to be saved, read, and updated whenever required.

Need for File Handling

- Stores data permanently.
- Reads previously saved info.
- Handles large amounts of data.
- Avoids repeated data entry.

Types of Files

↳ Text File

- Stores data in readable format.
- ex:- student.txt.
- Easy to create and edit.

Binary File

- stores data in binary format-
- Ex:- data.dat
- Faster and more memory - efficient.

File Pointer

↳ A File Pointer is used to perform file operations.

Declaration:

```
FILE *fp;
```

Here,

↳ FILE is a Predefined data type available in the stdio.h header file.

A File Pointer stores information about the opened file and keeps track of its current position.

without a File Pointer, file operations cannot be performed.

Opening a File

The fopen() function opens a file.

```
fp = fopen("student.txt", "r");
```

Always check whether the file is opened successfully.

```
if (fp == NULL) {
```

```
printf("File cannot be opened");
```

File Modes

Mode

Purpose

r	Read
w	Write
a	Append
r+	Read & write
w+	Read & write (overwrite)
a+	Read & Append

Common File Functions

Function

Purpose

fopen()	opens a file
fprintf()	writes formatted data
fscanf()	Reads formatted data
fgets()	Reads a complete line
fputc()	writes one character
fgetc()	Reads one character
fclose()	closes the file

Pointers in C

- ↳ A Pointer is a variable that stores the memory address of another variable.
- ↳ Instead of storing the value directly, it stores location where the value is stored in memory.

Example

```
int num = 25;
```

```
int *ptr = &num;
```

Here,

- ↳ num stores the value 25.
- ↳ &num gives the memory address of num.
- ↳ ptr stores the address.

Pointer operators

Address operator (&)

- ↳ Returns the memory address of a variable.
- ↳ `printf ("%p", &num);`

Dereference operator (*)

- ↳ Returns the value stored at the given address.
- ↳ `printf ("%d", *ptr);`

Pointer Example

```
#include <stdio.h>
```

```
int main ( )
```

```
{
```

```
int num = 50;
```

```
int *ptr = &num;
```

```
printf ("%d\n", *ptr);
```

```
return 0;
```

```
}
```

Output :-

50

Pointer vs File Pointer

Pointer

- ↳ stores the address of a variable.
- ↳ used for memory access
- ↳ declared as `int *p`, `char *p`, etc.
- ↳ works with variables.

File Pointer

- ↳ stores information about an opened file.
- ↳ used for file operations.
- ↳ declared as `FILE *fp`
- ↳ works with files.

↳ Applications

↳ File Handling

- Student Management System.
- Banking system.
- Hospital Records.
- Inventory Management.

↳ Pointers

- Dynamic Memory Allocation
- Arrays and strings
- Linked Lists
- Trees
- Graphs

Advantages

↳ File Handling

- Permanent storage of data.
- Easy retrieval of information.
- Efficient management of large datasets.
- Reduces repeated data entry.

b) Pointers

- Efficient memory usage.
- Faster Program execution.
- Supports dynamic memory allocation.
- Essential for implementing data structures.

Conclusion

b) File Handling is used to store and retrieve data permanently, while Pointers help access memory efficiently. Both are essential concepts in C programming and are widely used in developing efficient and reliable software.